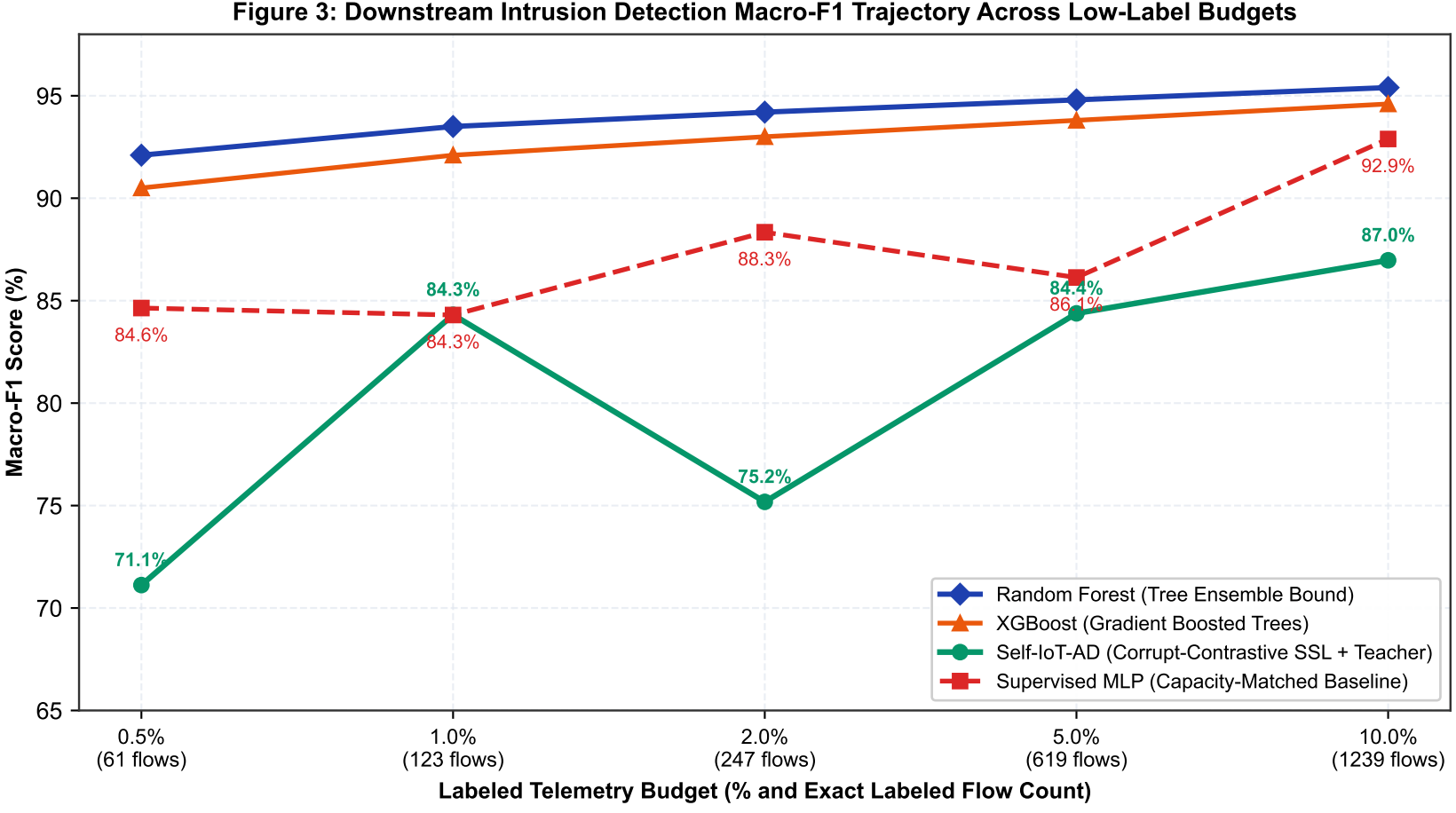
Figure 3: Downstream Intrusion Detection Macro-F1 Trajectory Across Low-Label Budgets
Macro-F1 Score (%)
95
90
85
80
75
70
65
0.5%
(61 flows)
1.0%
(123 flows)
2.0%
(247 flows)
5.0%
(619 flows)
10.0%
(1239 flows)
Labeled Telemetry Budget (% and Exact Labeled Flow Count)
84.6%
84.3%
88.3%
84.4%
92.9%
71.1%
84.3%
75.2%
86.1%
87.0%
Random Forest (Tree Ensemble Bound)
XGBoost (Gradient Boosted Trees)
Self-IoT-AD (Corrupt-Contrastive SSL + Teacher)
Supervised MLP (Capacity-Matched Baseline)